Supplementary Material

A Controlled Evaluation of Curvature-Inspired Scalar Feature Scaling for Domain Generalization

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1 Scope and Evidence Status

What the name refers to. HyperDG denotes the exact scalar feature-scaling formulation evaluated in the paper. It is not a citation to a separate external publication and it is not a name for hyperbolic domain generalization in general. The paper and this supplement specify the computation directly.

Claim boundary. The implementation performs scalar rescaling of Euclidean features during training and unscaled Euclidean linear prediction. It is not a Lorentz or Poincaré representation, a geodesic classifier, or a SAM inner loop. The evidence below therefore addresses the executed scalar package and the need for separate source-loader scalars. It does not address hyperbolic representation learning in general.

Table 1: Revision evidence inventory. Complete means every planned run reaches the end of the declared training schedule and contributes to the reported analysis.

Analysis	Status	Evidentiary use
Six-condition controlled grid	Complete: 384/384	Primary evidence: six conditions, four datasets, four targets, and four run seeds.
Native DomainBed ERM	Complete: 64/64	Whole-package utility comparator; not a curvature-only causal contrast because the update routes differ.
Shared-learnable source scale	Complete: 64/64	Direct domain-specificity comparison with NoVariance; Frozen is not used as its substitute.
Source-only model selection	Complete	Primary endpoint; held-out target accuracy is excluded from the frozen checkpoint score.
Target-clustered inference	Complete	Four targets are per-dataset clusters; four datasets are suite clusters.
Post-hoc Poincaré projections	Excluded	Visualization only; not evidence of manifold-valued training.
Legacy MERU attempts	Incomplete/excluded	Do not constitute a controlled genuine-hyperbolic baseline and support no accuracy claim.

2 Frozen Revision Protocol

2.1 Conditions

All HyperDG-path scale parameters (a_i and the global a_g; see Section 6) start at zero, so every condition begins with $c_i = c_g = \text{softplus}(0) + 10^{-4} = \log 2 + 10^{-4} = 0.6932472$, a feature multiplier of $1/\sqrt{c} = 1.2010358$. MatchedControl and Frozen set the scale optimizer's learning rate to zero, so a_i and a_g never move: $c_i = c_g = 0.6932472$ is the fixed value applied at *every* update of the run, not only at initialization. FullPerDomain, NoVariance, and SharedLearnable instead learn a_i (or, for SharedLearnable, one shared a) and a_g from this same zero starting point with the separate scale Adam optimizer described below. In a run with one held-out environment, per-domain means one scalar for each of three source-loader slots, not four persistent dataset-wide domain identities.

Only StandardERM, FullPerDomain, NoVariance, and SharedLearnable enter the paper's two primary suite contrasts (FullPerDomain−StandardERM and NoVariance−SharedLearnable, both defined in Section 2.5). MatchedControl and Frozen are secondary, descriptive diagnostics: no contrast reported in this paper treats either condition as a primary inferential claim. MatchedControl isolates what the HyperDG update route costs or gains at fixed, non-learned rescaling, before any auxiliary loss is active. Frozen is the fixed-scale counterpart of NoVariance, letting learned and fixed scaling be compared descriptively under the same non-dispersion auxiliary package; it is not a substitute for SharedLearnable, since it changes learnability rather than per-source specificity.

Table 2: Exact identities of the six conditions. Coefficients are $(\alpha_{\text{var}}, \beta_{\text{align}}, \gamma_{\text{norm}}, \delta_{\text{local}}, \delta_{\text{global}})$. Only the StandardERM, FullPerDomain, NoVariance, and SharedLearnable rows enter the two primary contrasts; MatchedControl and Frozen are descriptive.

Condition	Scales	Coefficients	Interpretation
StandardERM	None	n/a	Native concatenated-minibatch DomainBed ERM.
MatchedControl	Three local plus global, all fixed at 0.6932472	$(0, 0, 0, 0, 0)$	HyperDG route with auxiliary losses disabled; fixed feature rescaling remains. Descriptive only.
FullPerDomain	Three local plus global, learned	$(.2, .5, .05, .1, .1)$	Complete submitted implementation.
Frozen	Three local plus global, fixed at 0.6932472	$(0, .5, .05, .1, .1)$	Non-dispersion auxiliary package with fixed scales. Descriptive only.
NoVariance	Three local plus global, learned	$(0, .5, .05, .1, .1)$	Removes only explicit between-scale dispersion from FullPerDomain.
SharedLearnable	One shared local plus separate global, learned	$(.2, .5, .05, .1, .1)$	Dispersion is structurally zero despite $\alpha = .2$; clean comparator for NoVariance.

2.2 Data, targets, and training

Table 3: Named held-out targets. Every target is evaluated with seeds 0–3.

Dataset	Target 0	Target 1	Target 2	Target 3
PACS	ArtPainting	Cartoon	Photo	Sketch
VLCS	Caltech101	LabelMe	SUN09	VOC2007
OfficeHome	Art	Clipart	Product	RealWorld
TerraIncognita	L100	L38	L43	L46

All conditions use the same environment partitions and deterministic seed-dependent 80/20 splits, data augmentation, `timm` `resnet50.ram_in1k` ImageNet weights, 2,048-dimensional features, a biased linear classifier, batch size 32 per source, 5,000 updates, Adam network learning rate 5×10^{-5}, zero weight decay, and evaluation every 200 updates plus update 4999. Seeds change the split, initialization, minibatch order, and augmentation randomness. BatchNorm is active and dropout is zero. Learned scale parameters use Adam at 10^{-4}; the HyperDG route clips network and scale gradient norms to 5.

Native DomainBed ERM means the unmodified upstream DomainBed `ERM` class, run inside this project's identical data, split, augmentation, timm-ResNet-50, and source-only-selection harness: it concatenates the three source minibatches into one 96-example batch, computes one cross-entropy loss, and takes one Adam step with one optimizer and no gradient clipping. "Native" describes the update rule and objective, not a claim that the surrounding training/evaluation code is an untouched upstream DomainBed installation.

Per training step, the HyperDG route calls the feature extractor six times: three source-minibatch forward passes (with gradients) build the training objective in Equation (2) of the main paper, and, after the optimizer step, three further forward passes under `torch.no_grad()` recompute training accuracy for logging. `no_grad()` disables gradient tracking only; it does *not* switch the model to `eval()`, and the training loop never calls `eval()` during an update. All six forward passes therefore run with the model in ordinary training mode, and BatchNorm's running mean, running variance, and batch counter update exactly as they would for any other training-mode forward pass – including the three no-gradient diagnostic passes. Native ERM instead calls the feature extractor once per step, on the concatenated 96-example batch, so it updates BatchNorm statistics from one training-mode forward pass rather than HyperDG's six. This asymmetry, together with the separate scale optimizer and gradient clipping described above, is why FullPerDomain−StandardERM is a complete-package comparison rather than an isolated curvature effect. Both sides of NoVariance−SharedLearnable use the identical HyperDG route, so this asymmetry cancels out of that contrast.

2.3 Target-excluding checkpoint selection

For method m, dataset j, held-out target d, run seed s, source environment r, and checkpoint t, let V_{mjdsrt} be source `out`-split accuracy. The selected checkpoint is

$$t^*_{mjds} = \arg\max_t \frac{1}{3} \sum_{r \neq d} V_{mjdsrt}, \tag{1}$$

with earliest-checkpoint tie breaking. The held-out target key is not used in this calculation; its accuracy is read only after t^* is fixed. The trainer logged target values at every checkpoint, so we do not make the stronger claim that target curves never existed. Source validation inherits stochastic augmentation and is evaluated once per logged checkpoint.

2.4 Estimand

Every suite-level number in this supplement and the main paper is a claim about the equal-weighted macro over exactly PACS, VLCS, OfficeHome, and TerraIncognita under this protocol. It is not a claim about a population or superpopulation of DG tasks that these four benchmarks are sampled from: the suite was chosen for heterogeneous shift type (style, acquisition, context, and camera location; Section 2), not as a random draw from a larger task distribution. Figure 1 of the main paper states this explicitly because the suite-level interval, sign-flip test, and TOST below use the four dataset effects as inferential clusters without implying that conclusions transfer to DG tasks outside this fixed suite.

2.5 Estimands and inference

For a treatment A and comparator B, define

$$g_{jds} = Y_{Ajds} - Y_{Bjds}, \qquad \bar{g}_{jd} = \frac{1}{4} \sum_s g_{jds}, \qquad \Delta_j = \frac{1}{4} \sum_d \bar{g}_{jd}. \tag{2}$$

Seeds are averaged within target first ($\bar{g}_{jd}$), and targets are averaged within dataset second (Δ_j). The suite-level procedures below consume only the resulting four dataset means Δ_j; they do not separately repropagate within-target seed variance or between-target variance into the suite-level interval or test. Those lower-level components are real and are reported descriptively – per-target effects in Table 8, and the target/seed/residual decomposition in Table 9 – but they contribute to the suite-level uncertainty only through the four dataset means they produce, not as an additional variance term added at the suite level.

Inference within dataset j uses the four $\bar{g}_{jd}$ values, not the 16 seed-by-target cells. We enumerate all $2^4 = 16$ target-level sign assignments for an exact two-sided sign-flip test and separately report a t_3 interval. The sign-flip test is exact conditional on treating the four target-level signs as exchangeable under the joint no-effect null, i.e. that all 16 sign patterns are equally likely if each $\bar{g}_{jd}$ is symmetric about zero under the null. This is a working symmetry assumption suited to a four-cluster benchmark suite; it does not assume the datasets were randomly sampled or randomly assigned, and it is not a test of whether dataset-level effects cancel under a suite-macro-only null. The minimum attainable two-sided exact p-value is therefore .125 with four clusters. The suite macro averages the four Δ_j values and uses the four datasets as clusters. The two primary suite contrasts form a Holm-2 family; eight per-dataset contrasts are a separate sensitivity family.

We declare the exact sign-flip test with Holm correction as the primary inferential procedure for both suite-level difference claims. The t_3 interval and the four-cluster bootstrap sensitivity are secondary: they are working-model summaries that lean on approximate normality or resampling validity with only four clusters, and they do not override the primary result when the two disagree, as they do for FullPerDomain−StandardERM (Table 6: t_3 95% CI excludes zero, exact $p = .125$, Holm $p = .250$).

The revision-stage SESOI is $\pm 1\,pp$, selected after the originally submitted results were known but before the 128 added ERM/shared runs completed. Equivalence requires the clustered 90% interval to lie wholly inside that margin. Margins of ± 0.5 and $\pm 2\,pp$ are sensitivities. TOST, like the t_3 interval, is a secondary working-model summary conditional on the same fixed four-dataset macro (Section 2.4); failure to pass TOST is inconclusive, not evidence of nonequivalence, and a pass supports equivalence only for that fixed-suite macro.

3 Complete Source-Selected Results

3.1 Six-condition summary with separated variability

Table 4 separates between-target SD from the root-mean square within-target run-seed SD. The latter combines split, initialization, minibatch-order, and augmentation randomness; it is not a pure optimization seed SD.

Table 4: Source-selected accuracy and two distinct variability components. Accuracy is the equal-target macro mean.

Condition	Dataset	Accuracy (%)	Between-target SD	Within-target run-seed SD (RMS)
Standard ERM	PACS	85.88	7.98	3.04
Standard ERM	VLCS	77.19	14.09	1.49
Standard ERM	OfficeHome	70.21	10.90	1.79
Standard ERM	TerraIncognita	47.13	6.62	4.33
Matched control	PACS	84.22	7.99	3.39
Matched control	VLCS	75.38	15.07	2.44
Matched control	OfficeHome	70.05	10.73	2.31
Matched control	TerraIncognita	44.14	8.70	4.21
Full per-domain	PACS	84.87	8.19	1.89
Full per-domain	VLCS	76.42	14.90	1.65
Full per-domain	OfficeHome	68.52	10.33	2.54
Full per-domain	TerraIncognita	44.17	4.60	4.28
Frozen	PACS	86.14	7.22	2.19
Frozen	VLCS	75.84	14.30	2.22
Frozen	OfficeHome	67.81	11.91	1.57
Frozen	TerraIncognita	44.20	6.98	3.41
No-variance per-domain	PACS	85.28	7.76	2.00
No-variance per-domain	VLCS	76.17	14.95	1.19
No-variance per-domain	OfficeHome	68.36	10.10	2.64
No-variance per-domain	TerraIncognita	43.70	7.90	4.72
Shared learnable	PACS	85.55	7.54	3.01
Shared learnable	VLCS	76.56	14.65	2.20
Shared learnable	OfficeHome	68.17	11.15	2.09
Shared learnable	TerraIncognita	43.75	5.35	4.73

Across the 24 condition-by-dataset cells, between-target SD ranges from 4.60 to 15.07 *pp*, while within-target run-seed SD (RMS) ranges from 1.19 to 4.73 *pp*. Pooling these sources into one SD would obscure the dominant target-difficulty variation in several datasets.

3.2 Every target and all four run seeds

Table 5: Every source-selected held-out-target accuracy, with all four run seeds, target mean, and run-seed SD.

Condition	Dataset	Target	Seed 0	Seed 1	Seed 2	Seed 3	Mean	Run-seed SD
Standard ERM	PACS	ArtPainting	83.62	91.93	81.17	86.80	85.88	4.64
Standard ERM	PACS	Cartoon	82.69	81.62	79.70	75.43	79.86	3.20
Standard ERM	PACS	Photo	99.40	97.01	97.01	95.21	97.16	1.72
Standard ERM	PACS	Sketch	82.17	78.73	81.27	80.38	80.64	1.47
Standard ERM	VLCS	Caltech101	98.59	96.11	95.76	97.17	96.91	1.27
Standard ERM	VLCS	LabelMe	66.42	63.21	64.53	62.83	64.25	1.62
Standard ERM	VLCS	SUN09	71.19	69.36	71.34	71.80	70.92	1.07
Standard ERM	VLCS	VOC2007	78.52	75.70	77.93	74.52	76.67	1.88
Standard ERM	OfficeHome	Art	67.01	64.54	64.54	66.39	65.62	1.28
Standard ERM	OfficeHome	Clipart	57.16	53.15	58.53	59.11	56.99	2.69
Standard ERM	OfficeHome	Product	79.93	78.58	77.00	77.11	78.16	1.38
Standard ERM	OfficeHome	RealWorld	80.25	78.19	80.25	81.63	80.08	1.42
Standard ERM	TerraIncognita	L100	45.85	47.38	46.15	57.33	49.18	5.48
Standard ERM	TerraIncognita	L38	48.08	49.31	42.45	36.05	43.97	6.07
Standard ERM	TerraIncognita	L43	56.09	54.98	53.11	57.09	55.32	1.71
Standard ERM	TerraIncognita	L46	42.32	40.20	36.93	40.77	40.05	2.27
Matched control	PACS	ArtPainting	86.06	86.55	82.15	83.86	84.66	2.04
Matched control	PACS	Cartoon	81.84	79.70	77.14	73.93	78.15	3.41
Matched control	PACS	Photo	96.11	93.71	95.81	95.81	95.36	1.11
Matched control	PACS	Sketch	83.18	82.93	71.97	76.69	78.69	5.39
Matched control	VLCS	Caltech101	95.41	96.11	96.11	98.23	96.47	1.22
Matched control	VLCS	LabelMe	60.75	61.13	59.62	61.32	60.71	0.76
Matched control	VLCS	SUN09	74.09	70.58	72.87	72.87	72.60	1.46
Matched control	VLCS	VOC2007	74.96	75.41	70.67	65.93	71.74	4.43
Matched control	OfficeHome	Art	64.12	65.15	62.27	69.69	65.31	3.16
Matched control	OfficeHome	Clipart	59.11	53.61	56.47	59.34	57.13	2.68
Matched control	OfficeHome	Product	78.80	80.27	78.92	76.44	78.61	1.59
Matched control	OfficeHome	RealWorld	78.65	77.61	80.02	80.37	79.16	1.27
Matched control	TerraIncognita	L100	58.46	49.23	56.51	57.64	55.46	4.23
Matched control	TerraIncognita	L38	42.55	47.31	42.45	33.44	41.44	5.80
Matched control	TerraIncognita	L43	42.66	50.37	43.53	43.66	45.06	3.57
Matched control	TerraIncognita	L46	38.07	35.13	32.19	33.09	34.62	2.61
Full per-domain	PACS	ArtPainting	84.11	85.33	83.62	83.86	84.23	0.76
Full per-domain	PACS	Cartoon	78.85	78.85	76.28	76.71	77.67	1.37
Full per-domain	PACS	Photo	95.51	97.31	96.71	96.41	96.48	0.75
Full per-domain	PACS	Sketch	81.40	85.61	79.75	77.71	81.11	3.35
Full per-domain	VLCS	Caltech101	99.29	96.47	97.17	96.82	97.44	1.27
Full per-domain	VLCS	LabelMe	62.08	62.26	62.83	62.08	62.31	0.36
Full per-domain	VLCS	SUN09	73.32	69.82	73.93	71.65	72.18	1.85
Full per-domain	VLCS	VOC2007	75.70	75.41	73.33	70.52	73.74	2.39
Full per-domain	OfficeHome	Art	65.36	62.27	56.91	65.36	62.47	3.99
Full per-domain	OfficeHome	Clipart	55.90	55.21	57.96	59.22	57.07	1.85
Full per-domain	OfficeHome	Product	79.26	79.14	74.75	76.21	77.34	2.23
Full per-domain	OfficeHome	RealWorld	76.46	76.00	78.87	77.38	77.18	1.27
Full per-domain	TerraIncognita	L100	48.00	44.82	55.28	48.31	49.10	4.41
Full per-domain	TerraIncognita	L38	34.10	44.91	43.16	39.27	40.36	4.79
Full per-domain	TerraIncognita	L43	40.42	49.50	48.88	49.50	47.08	4.45
Full per-domain	TerraIncognita	L46	42.65	35.62	39.62	42.65	40.13	3.33
Frozen	PACS	ArtPainting	88.26	88.51	84.11	83.86	86.19	2.54
Frozen	PACS	Cartoon	84.19	77.78	77.35	79.06	79.59	3.15
Frozen	PACS	Photo	95.81	95.51	96.71	96.71	96.18	0.62
Frozen	PACS	Sketch	82.17	84.59	82.68	80.89	82.58	1.53
Frozen	VLCS	Caltech101	98.23	94.70	96.82	95.05	96.20	1.64
Frozen	VLCS	LabelMe	62.83	65.66	61.13	61.32	62.74	2.09
Frozen	VLCS	SUN09	74.54	68.45	74.70	73.32	72.75	2.94
Frozen	VLCS	VOC2007	70.37	74.22	72.30	69.78	71.67	2.01
Frozen	OfficeHome	Art	62.06	61.03	60.82	63.30	61.80	1.13
Frozen	OfficeHome	Clipart	53.95	52.00	55.33	54.64	53.98	1.43
Frozen	OfficeHome	Product	78.92	75.99	77.00	76.44	77.09	1.29

Condition	Dataset	Target	Seed 0	Seed 1	Seed 2	Seed 3	Mean	Run-seed SD
Frozen	OfficeHome	RealWorld	76.23	76.81	79.79	80.71	78.39	2.20
Frozen	TerraIncognita	L100	55.18	46.67	49.33	57.23	52.10	4.93
Frozen	TerraIncognita	L38	37.79	42.75	40.91	37.79	39.81	2.45
Frozen	TerraIncognita	L43	43.66	45.77	50.00	51.99	47.85	3.82
Frozen	TerraIncognita	L46	36.36	36.52	36.44	38.89	37.05	1.23
No-variance per-domain	PACS	ArtPainting	83.62	82.89	87.53	82.89	84.23	2.23
No-variance per-domain	PACS	Cartoon	79.27	77.56	77.56	77.99	78.10	0.81
No-variance per-domain	PACS	Photo	97.60	94.61	95.81	97.01	96.26	1.33
No-variance per-domain	PACS	Sketch	84.71	85.22	79.11	81.15	82.55	2.92
No-variance per-domain	VLCS	Caltech101	97.88	97.17	96.82	98.23	97.53	0.65
No-variance per-domain	VLCS	LabelMe	63.02	63.21	63.96	60.38	62.64	1.56
No-variance per-domain	VLCS	SUN09	72.26	71.04	72.56	70.73	71.65	0.90
No-variance per-domain	VLCS	VOC2007	73.63	73.93	73.04	70.81	72.85	1.41
No-variance per-domain	OfficeHome	Art	60.62	59.38	68.25	60.41	62.16	4.09
No-variance per-domain	OfficeHome	Clipart	57.27	54.07	56.82	61.40	57.39	3.02
No-variance per-domain	OfficeHome	Product	76.89	78.69	75.99	75.99	76.89	1.28
No-variance per-domain	OfficeHome	RealWorld	77.04	76.35	76.92	77.61	76.98	0.52
No-variance per-domain	TerraIncognita	L100	60.10	52.41	51.59	46.77	52.72	5.52
No-variance per-domain	TerraIncognita	L38	39.22	35.28	44.55	33.08	38.03	5.03
No-variance per-domain	TerraIncognita	L43	49.25	43.66	47.39	51.12	47.85	3.19
No-variance per-domain	TerraIncognita	L46	43.06	32.68	35.95	33.09	36.19	4.80
Shared learnable	PACS	ArtPainting	80.68	88.26	84.84	87.04	85.21	3.33
Shared learnable	PACS	Cartoon	83.33	79.91	75.00	77.35	78.90	3.57
Shared learnable	PACS	Photo	97.31	97.31	95.81	94.31	96.18	1.43
Shared learnable	PACS	Sketch	84.08	84.84	77.83	80.89	81.91	3.21
Shared learnable	VLCS	Caltech101	97.88	97.17	96.11	97.88	97.26	0.83
Shared learnable	VLCS	LabelMe	65.85	59.81	64.15	61.51	62.83	2.69
Shared learnable	VLCS	SUN09	73.02	68.75	74.39	71.65	71.95	2.41
Shared learnable	VLCS	VOC2007	76.59	75.85	72.30	72.00	74.19	2.37
Shared learnable	OfficeHome	Art	61.03	59.18	63.09	64.95	62.06	2.50
Shared learnable	OfficeHome	Clipart	57.04	52.46	53.84	58.76	55.53	2.89
Shared learnable	OfficeHome	Product	78.58	79.26	76.10	76.78	77.68	1.48
Shared learnable	OfficeHome	RealWorld	76.46	78.19	77.15	77.84	77.41	0.76
Shared learnable	TerraIncognita	L100	57.54	46.77	49.23	45.03	49.64	5.54
Shared learnable	TerraIncognita	L38	44.75	47.52	38.76	33.69	41.18	6.19
Shared learnable	TerraIncognita	L43	45.15	41.79	50.50	48.63	46.52	3.85
Shared learnable	TerraIncognita	L46	40.77	34.97	37.99	36.93	37.66	2.42

The preceding table is the complete 384-cell accuracy inventory: six conditions, four datasets, four named targets, and four seeds. Values are the held-out target `out`-split accuracies at the source-selected checkpoint. No target or seed is omitted.

4 Primary Contrasts and Small-Cluster Inference

4.1 Suite and per-dataset estimates

Table 6: Primary suite-macro analyses. Exact tests flip the four dataset effects; Holm adjusts the two exact p-values. TOST uses the revision-stage $\pm 1\,pp$ margin.

Contrast	Effect	95% CI	Exact p	Holm p	90% CI	TOST p
FullPerDomain−StandardERM	−1.61	$[-3.17, -0.04]$	.125	.250	$[-2.77, -0.45]$	.848
NoVariance−SharedLearnable	−0.13	$[-0.53, 0.27]$	.375	.375	$[-0.43, 0.17]$	.003

Table 7: Per-dataset target-clustered sensitivities. Holm adjusts these eight exact tests as a sensitivity family. No row establishes equivalence within $\pm 1\,pp$.

Contrast	Dataset	Effect	95% CI	Exact p	Holm p	90% CI	TOST p
FullPerDomain−StandardERM	PACS	−1.01	$[-2.88, 0.86]$	.250	1.000	$[-2.39, 0.37]$	.506
	VLCS	−0.77	$[-3.92, 2.39]$	.500	1.000	$[-3.10, 1.57]$	.415
	OfficeHome	−1.69	$[-4.21, 0.82]$	.250	1.000	$[-3.55, 0.17]$	.778
	TerraIncognita	−2.96	$[-9.18, 3.26]$	.375	1.000	$[-7.56, 1.64]$	.805
NoVariance−SharedLearnable	PACS	−0.27	$[-1.47, 0.94]$	.500	1.000	$[-1.16, 0.63]$	.074
	VLCS	−0.39	$[-1.46, 0.68]$	.375	1.000	$[-1.18, 0.40]$	.084
	OfficeHome	0.19	$[-1.68, 2.06]$	.875	1.000	$[-1.20, 1.57]$	.130
	TerraIncognita	−0.05	$[-4.49, 4.39]$	.875	1.000	$[-3.33, 3.23]$	.273

The suite t_3 interval for FullPerDomain−StandardERM excludes zero, while the exact four-dataset test is $p = .125$ and Holm-adjusted $p = .250$. We retain both results: the parametric interval summarizes compatible cross-dataset effects, whereas the exact test exposes the resolution of four clusters. We do not claim an adjusted difference. All four dataset point estimates favor ERM, and equivalence of the packages is not demonstrated.

For NoVariance−SharedLearnable, the source-selected suite macro is $-0.13\,pp$ with 90% CI $[-0.43, 0.17]$. It passes the declared $\pm 1\,pp$ TOST ($p = .003$). No individual dataset passes a $\pm 1\,pp$ equivalence test, so the conclusion is restricted to the equal-dataset/equal-target macro estimand.

4.2 Named-target effects

Table 8: Mean paired effects after averaging the four paired run-seed differences within target. Run-seed-pair SD is descriptive.

Contrast	Dataset	Target	Mean paired effect (pp)	Run-seed-pair SD
Full per-domain - Standard ERM	PACS	ArtPainting	-1.65	3.98
Full per-domain - Standard ERM	PACS	Cartoon	-2.19	2.36
Full per-domain - Standard ERM	PACS	Photo	-0.67	2.23
Full per-domain - Standard ERM	PACS	Sketch	0.48	4.34
Full per-domain - Standard ERM	VLCS	Caltech101	0.53	0.74
Full per-domain - Standard ERM	VLCS	LabelMe	-1.93	1.65
Full per-domain - Standard ERM	VLCS	SUN09	1.26	1.31
Full per-domain - Standard ERM	VLCS	VOC2007	-2.93	1.90
Full per-domain - Standard ERM	OfficeHome	Art	-3.14	3.03
Full per-domain - Standard ERM	OfficeHome	Clipart	0.09	1.43
Full per-domain - Standard ERM	OfficeHome	Product	-0.82	1.15
Full per-domain - Standard ERM	OfficeHome	RealWorld	-2.90	1.35
Full per-domain - Standard ERM	TerraIncognita	L100	-0.08	7.66
Full per-domain - Standard ERM	TerraIncognita	L38	-3.61	7.61
Full per-domain - Standard ERM	TerraIncognita	L43	-8.24	5.14
Full per-domain - Standard ERM	TerraIncognita	L46	0.08	3.26
No-variance per-domain - Shared learnable	PACS	ArtPainting	-0.98	4.41
No-variance per-domain - Shared learnable	PACS	Cartoon	-0.80	2.97
No-variance per-domain - Shared learnable	PACS	Photo	0.07	2.21
No-variance per-domain - Shared learnable	PACS	Sketch	0.64	0.45
No-variance per-domain - Shared learnable	VLCS	Caltech101	0.27	0.34
No-variance per-domain - Shared learnable	VLCS	LabelMe	-0.19	2.63
No-variance per-domain - Shared learnable	VLCS	SUN09	-0.30	1.79
No-variance per-domain - Shared learnable	VLCS	VOC2007	-1.33	1.56
No-variance per-domain - Shared learnable	OfficeHome	Art	0.10	3.97
No-variance per-domain - Shared learnable	OfficeHome	Clipart	1.86	1.24
No-variance per-domain - Shared learnable	OfficeHome	Product	-0.79	0.66
No-variance per-domain - Shared learnable	OfficeHome	RealWorld	-0.43	1.01
No-variance per-domain - Shared learnable	TerraIncognita	L100	3.08	1.74
No-variance per-domain - Shared learnable	TerraIncognita	L38	-3.15	7.63
No-variance per-domain - Shared learnable	TerraIncognita	L43	1.34	3.11
No-variance per-domain - Shared learnable	TerraIncognita	L46	-1.47	2.63

4.3 Paired-effect variability decomposition

Table 9: Descriptive decomposition of each primary paired effect. It keeps target, run-seed, and interaction variation distinct; it is not an additional inferential test.

Dataset	Mean effect	Target SD	Seed SD	Within-target seed SD	Residual SD
FullPerDomain−StandardERM					
PACS	−1.01	1.17	0.67	3.36	3.80
VLCS	−0.77	1.98	0.54	1.47	1.58
OfficeHome	−1.69	1.58	1.03	1.90	1.84
TerraIncognita	−2.96	3.91	3.73	6.20	5.71
NoVariance−SharedLearnable					
PACS	−0.27	0.76	1.70	2.88	2.69
VLCS	−0.39	0.67	1.08	1.78	1.63
OfficeHome	0.19	1.18	1.20	2.17	2.08
TerraIncognita	−0.05	2.79	1.21	4.41	4.90

4.4 Equivalence-margin sensitivity

Table 10: Suite-macro TOST sensitivity at ±0.5, ±1, and ±2 *pp*. Only NoVariance−SharedLearnable at ±1 *pp* is the declared revision-stage equivalence analysis; the other rows are sensitivities. Dataset-level ±1 *pp* results are in Table 7.

Contrast	Margin	Effect	90% CI	TOST p	Conclusion
FullPerDomain−StandardERM	±0.5	−1.61	[−2.77, −0.45]	.945	Not demonstrated
	±1.0	−1.61	[−2.77, −0.45]	.848	Not demonstrated
	±2.0	−1.61	[−2.77, −0.45]	.242	Not demonstrated
NoVariance−SharedLearnable	±0.5	−0.13	[−0.43, 0.17]	.031	Equivalent
	±1.0	−0.13	[−0.43, 0.17]	.003	Equivalent
	±2.0	−0.13	[−0.43, 0.17]	< .001	Equivalent

5 Checkpoint-Selection Sensitivity

Table 11: Source-selected versus update-4999 target accuracy. Selection uses only mean source `out` accuracy.

Condition	Dataset	Source-selected (%)	Final-step (%)	Selected - final (pp)
Standard ERM	PACS	85.88	85.28	0.60
Standard ERM	VLCS	77.19	75.72	1.46
Standard ERM	OfficeHome	70.21	67.47	2.75
Standard ERM	TerraIncognita	47.13	46.57	0.56
Matched control	PACS	84.22	84.04	0.17
Matched control	VLCS	75.38	75.63	-0.25
Matched control	OfficeHome	70.05	67.05	3.00
Matched control	TerraIncognita	44.14	43.90	0.24
Full per-domain	PACS	84.87	84.50	0.38
Full per-domain	VLCS	76.42	75.47	0.95
Full per-domain	OfficeHome	68.52	66.03	2.49
Full per-domain	TerraIncognita	44.17	43.17	1.00
Frozen	PACS	86.14	84.67	1.47
Frozen	VLCS	75.84	76.27	-0.43
Frozen	OfficeHome	67.81	66.13	1.68
Frozen	TerraIncognita	44.20	44.30	-0.10
No-variance per-domain	PACS	85.28	84.57	0.71
No-variance per-domain	VLCS	76.17	75.04	1.13
No-variance per-domain	OfficeHome	68.36	66.03	2.33
No-variance per-domain	TerraIncognita	43.70	43.90	-0.20
Shared learnable	PACS	85.55	85.10	0.45
Shared learnable	VLCS	76.56	76.12	0.44
Shared learnable	OfficeHome	68.17	66.27	1.90
Shared learnable	TerraIncognita	43.75	43.32	0.43

Source selection exceeds final-step reporting in 20 of 24 condition-by-dataset cells, with an average change of +0.97 *pp*. At update 4999 the NoVariance−SharedLearnable suite effect is −0.32 *pp* with 90% CI [−1.13, 0.50] and TOST $p = .072$. Thus final-step equivalence is not established, and the paper's equivalence claim is explicitly endpoint-specific.

5.1 Held-out target in-split sensitivity

Table 12: Source-selected held-out target `in`-split sensitivity. These values never enter checkpoint selection.

Contrast	Dataset	Target-in effect (pp)	95% CI	Exact p
Full per-domain - Standard ERM	PACS	-0.64	[-2.12, 0.84]	0.375
Full per-domain - Standard ERM	VLCS	-0.98	[-3.88, 1.92]	0.375
Full per-domain - Standard ERM	OfficeHome	-1.56	[-3.84, 0.72]	0.250
Full per-domain - Standard ERM	TerraIncognita	-2.49	[-7.33, 2.35]	0.250
No-variance per-domain - Shared learnable	PACS	-0.25	[-3.09, 2.59]	0.875
No-variance per-domain - Shared learnable	VLCS	-0.51	[-1.68, 0.66]	0.250
No-variance per-domain - Shared learnable	OfficeHome	0.17	[-1.52, 1.85]	1.000
No-variance per-domain - Shared learnable	TerraIncognita	0.53	[-3.83, 4.89]	0.875

6 Implementation Audit

6.1 Scalar training path and unscaled prediction

For local source slot i, the update path uses

$$c_i = \mathrm{softplus}(a_i) + 10^{-4}, \qquad z_i = h_\theta(x_i)/\sqrt{c_i}, \tag{3}$$

and the supervised loss classifies z_i. A separate scalar c_g is used for the global disagreement term. By contrast, the prediction method is equivalent to

$$\mathrm{predict}(x) = W h_\theta(x) + b. \tag{4}$$

No local or global scale is applied to target inference or to logged checkpoint accuracy. The learned values can influence training gradients, but they do not define a target-domain hyperbolic prediction rule.

6.2 Alignment-loss weights

The domain-mean alignment term in the main paper's Equation (3) is

$$\mathcal{L}_{\mathrm{align}} = \frac{1}{3}\sum_i \left\| \bar{z}_i - \sum_j w_j \bar{z}_j \right\|_2^2, \tag{5}$$

where $\bar{z}_i = \frac{1}{|B_i|}\sum_{k\in B_i} z_{i,k}$ averages the scaled features over the $|B_i| = 32$ examples in source minibatch i, and $w_j = B_j/(\sum_k B_k + 10^{-8})$ weights each domain mean by its minibatch size (the 10^{-8} term is a numerical-stability offset in the denominator). Every source minibatch has 32 examples in this protocol, so $w_j = 1/3$ for all three source slots and $\sum_j w_j \bar{z}_j$ is simply the unweighted mean of the three domain means.

6.3 Inactive local-gradient regularizer

The forward pass indexes the transformed vector of scales and uses that view to construct z_i. The local diagnostic separately indexes the underlying parameter vector and asks for a gradient with respect to the new view. That view is not the tensor used in the forward graph. With `allow_unused=True`, automatic differentiation returns `None`; the implementation converts it to zero. Consequently,

$$\mathcal{L}_{\mathrm{local}}^{\mathrm{executed}} = 0 \tag{6}$$

despite the configured coefficient $\delta_{\mathrm{local}} = .1$ in the non-control conditions. The logged local term is identically zero in every relevant completed run. This is an implementation defect and is disclosed rather than silently repaired only for the added arms.

6.4 Algebraic cancellation in the norm proxy

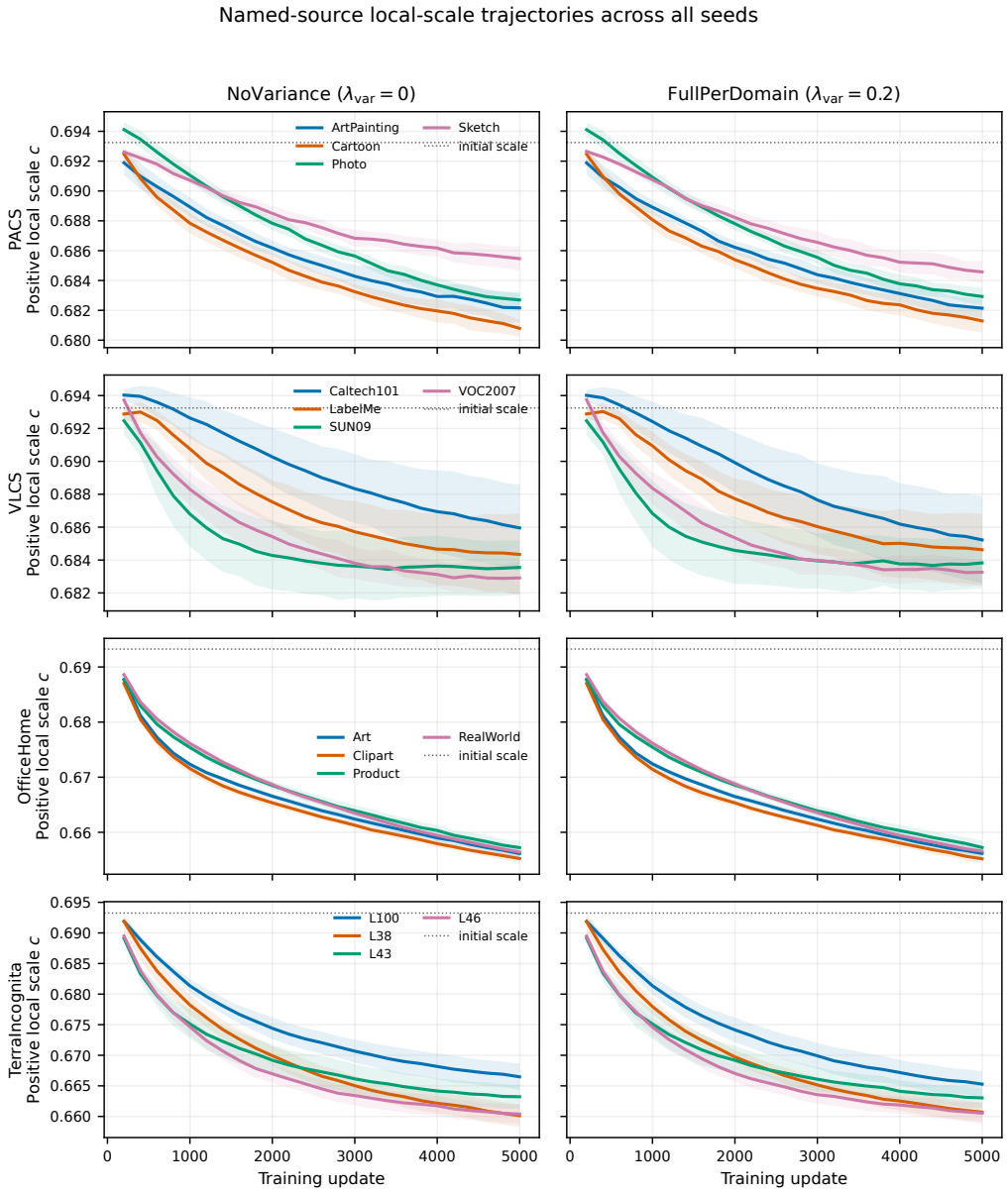
The implementation computes

$$\mathcal{L}_{\mathrm{norm},i} = c_i \ \mathrm{Var}(\|h_i/\sqrt{c_i}\|_2) = \mathrm{Var}(\|h_i\|_2), \tag{7}$$

apart from an inactive clamp at the tested values, where $\mathrm{Var}(\cdot)$ is the population variance (divisor $|B_i| = 32$, PyTorch `unbiased=False`) of the 32 per-example scaled-feature norms $\|h_{i,k}/\sqrt{c_i}\|_2$, $k \in B_i$, taken within source minibatch i before the three per-domain values are averaged. Because $\sqrt{c_i}$ is the same constant for every example in minibatch i, it factors out of the variance and the multiplying c_i cancels it exactly, leaving $\mathrm{Var}(\|h_i\|_2)$. The term can regularize the encoder through feature gradients, but its value does not depend on c_i, so it cannot identify c_i and is not a curvature sharpness quantity. It also performs no parameter perturbation or inner maximization and therefore is not SAM.

6.5 Source-loader indexing

With target environment d held out, loader slots enumerate the remaining environment indices in sorted order. For example, PACS target Cartoon removes environment 1, so slots 0, 1, and 2 correspond to ArtPainting, Photo, and Sketch. When a different target is held out, that mapping changes. A trajectory label such as slot 1 therefore cannot be treated as one semantic domain across all leave-one-domain-out runs. The revised trajectory rendering writes the named source mapping in every panel and summarizes all four seeds.



Loader slot j is remapped to the j-th original environment after removing the held-out target. Lines and bands are descriptive mean ± SD over 3 target contexts × 4 seeds (n=12; overlapping source data).

Figure 1: Learned local-scale trajectories for FullPerDomain and NoVariance, aggregated over all four run seeds and labeled by the named source represented by each loader slot. Bands are descriptive mean ± SD over 12 overlapping target-context/run-seed observations per named source and checkpoint, not independent confidence intervals. The plotted positive values are Euclidean feature scales, not learned sectional curvatures.

Table 13: Final within-run range across the three learned local scales. Each cell gives the mean range over 16 target/seed runs followed by the maximum range. This all-seed summary replaces the earlier seed-0-only diagnostic.

Dataset	NoVariance mean / max	FullPerDomain mean / max
PACS	0.003837 / 0.005934	0.002667 / 0.004499
VLCS	0.003169 / 0.006600	0.002475 / 0.004947
OfficeHome	0.001482 / 0.002970	0.001445 / 0.002477
TerraIncognita	0.006020 / 0.010730	0.004476 / 0.008534

6.6 SharedLearnable and global-scale trajectories

Figure 1 above shows only FullPerDomain and NoVariance, since those are the two conditions with three named per-source local scales. SharedLearnable has a single learned local scale by construction, so it cannot be decomposed by named source; to compare it against NoVariance on the same axis, we summarize NoVariance's three per-run local scales by their per-run mean. This mean is a descriptive summary constructed only for this comparison, not a new inferential estimand.

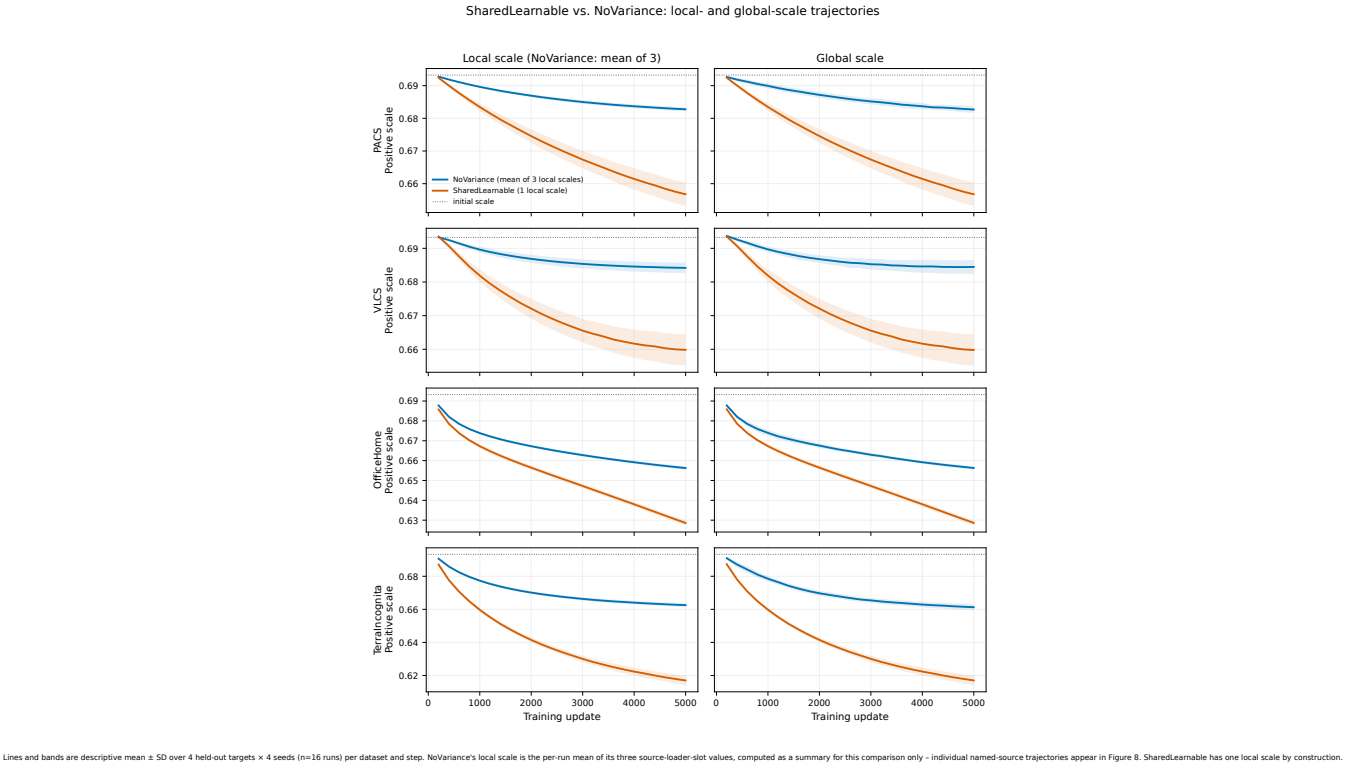


Figure 2: Local-scale (NoVariance's per-run mean of three source-loader-slot values vs. SharedLearnable's single value) and global-scale trajectories, aggregated over all four held-out targets and four run seeds (mean ± SD, $n = 16$ runs per dataset and checkpoint). Both conditions start every scale at 0.693247 (dotted line). Descriptively, SharedLearnable's single local and global scales move further from this initial value than NoVariance's per-run mean local scale does, on all four datasets; we do not treat this as evidence for or against either parameterization; it is not one of the paper's two primary contrasts, and Table 6 already reports the performance-based NoVariance−SharedLearnable equivalence result that is this comparison's inferential counterpart.

6.7 BatchNorm and optimizer routes

BatchNorm is not frozen. The HyperDG update processes the three source minibatches separately, then performs an additional no-gradient prediction pass for a training diagnostic while remaining in train mode. Native ERM concatenates source minibatches and performs one feature-extractor pass. The routes therefore update running statistics differently. HyperDG also uses separate Adam optimizers for network and scale parameters and clips both gradient norms; native ERM retains its unmodified single optimizer without that clipping. These differences are part of the treatment in FullPerDomain−StandardERM, which is why the paper calls it a package-utility comparison.

7 Learning-Curve Diagnostics

Target curves are generated after training for diagnosis and never enter the selection function. Cross markers indicate checkpoints selected solely by mean stochastic source-validation accuracy.

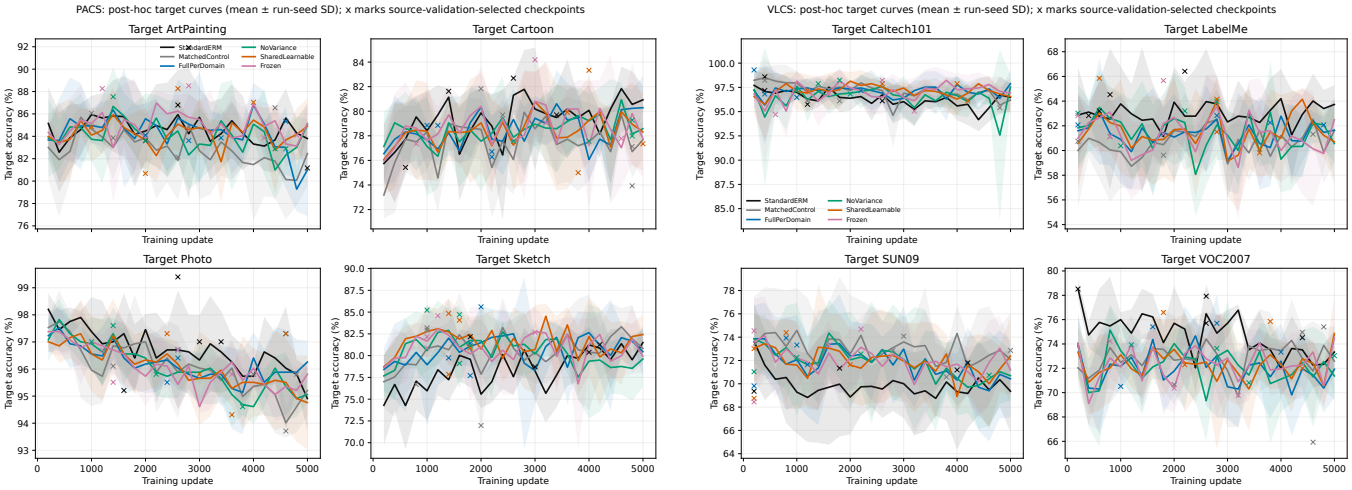


Figure 3: PACS and VLCS learning curves by named held-out target. Curves are descriptive; checkpoint selection uses only source-validation accuracy.

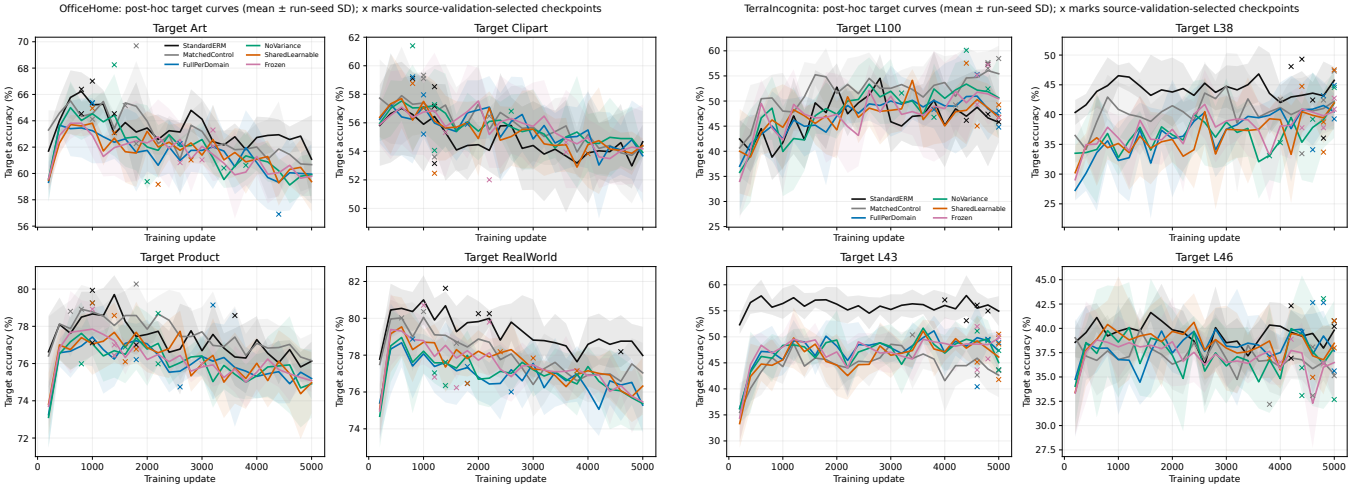


Figure 4: OfficeHome and TerraIncognita learning curves by named held-out target.

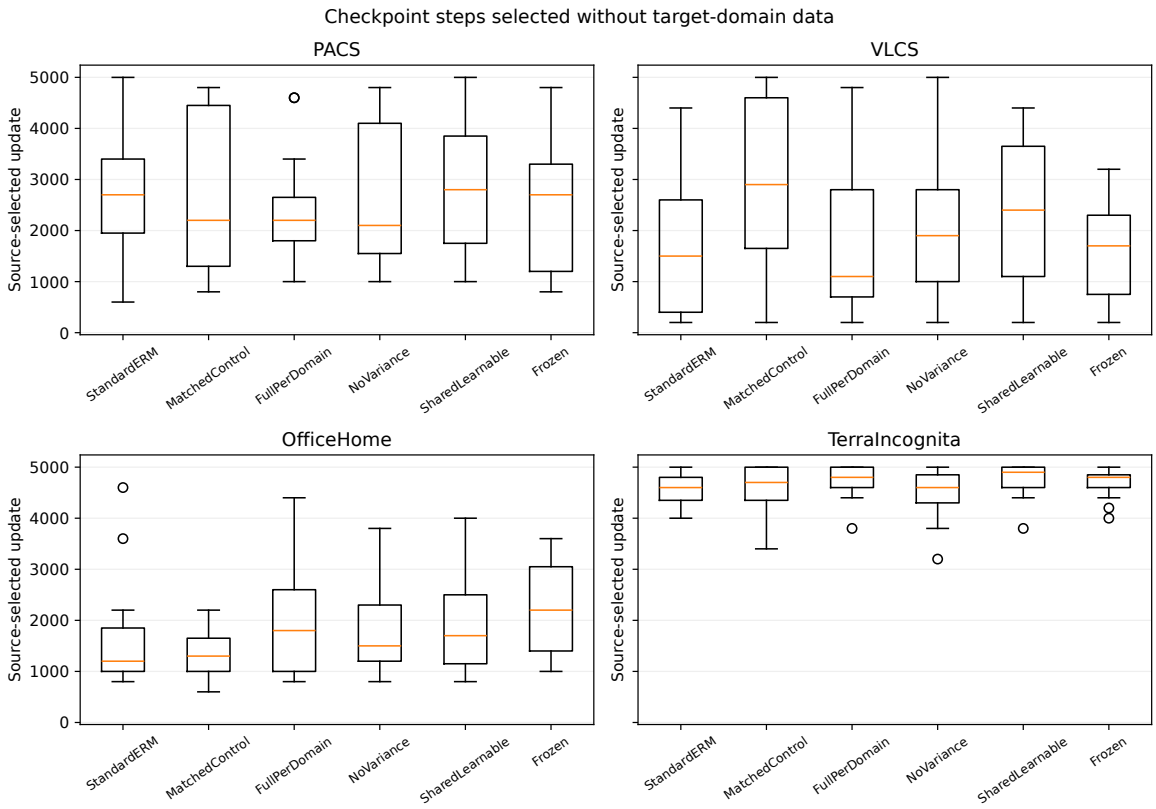


Figure 5: Distribution of source-selected checkpoint steps by condition and dataset.

8 Why the Existing Geometry Diagnostics Are Excluded

Table 14: Disposition of geometry-related analyses outside the 384-run controlled matrix.

Analysis	What it computes	Why it supports no main claim
Post-hoc Poincaré projection	Maps already trained Euclidean features for a two-dimensional visualization	Does not train a hyperbolic encoder or classifier; projection appearance cannot validate learned curvature.
Legacy MERU adaptation	Partial attempt to add a Lorentz-style comparison	Run grid is incomplete and not protocol-matched; it cannot answer the reviewer's genuine-hyperbolic request.
Old seed-0 shared runs	One shared scalar in a mixed software environment	Incomplete and environment-mismatched; replaced by all 64 fresh SharedLearnable runs.
Frozen scales	Fixed scalars with other auxiliary terms active	Tests learnability, not one learned shared scale; not used for domain specificity.

A valid genuine-hyperbolic extension would implement a manifold-valued Poincaré or Lorentz representation and appropriate classifier under the same splits, seeds, backbone, budget, and target-excluding selector. It would need fixed- and learned-curvature controls and the same target-clustered analysis. No result in the current study meets that definition, so the paper makes no claim about such methods.